

Summmer Project: Double Pendulum

Siddharth Matia

Under supervision of Christian Hölbling
Department of Physics, Wuppertal University Germany

September 25, 2023

1 Methods of integration

This project focuses on solving ordinary differential equations numerically. We can solve differential equations following several methods. We study three different methods: the explicit Euler method, and the Runge-Kutta 2 and 4 methods.

1.1 Euler method

The first method, the Euler method, is one of the most basic forms of numerical integration. It works as follows.

1. Choose a domain on which to solve the differential equation on.
2. Split that domain up into small intervals. That interval is called Δt . A smaller Δt will mean more accurate results, but more computation. Note: The actual accuracy is dependent on the differential equation itself.
3. Find the slope at a point using the given derivative. For the first step, the point will be the start of the domain chosen in the first step.
4. Use that slope to travel for a step size of Δt
5. Repeat steps 2 and forward

This gives a base level approximation of the integration of some inputted function. The result of this algorithm for the differential equation $\dot{x} = e^t$ is shown below (for a Δt of 1).

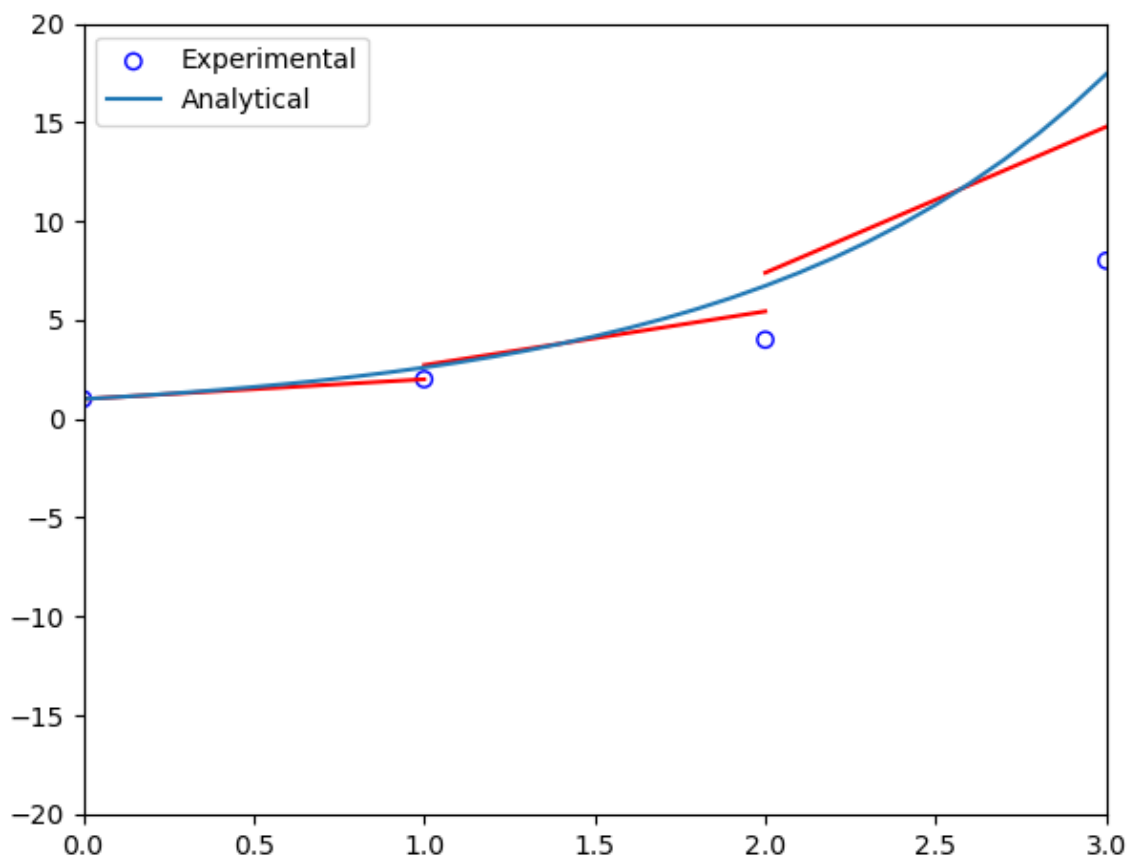


Figure 1: Euler method estimate of $\dot{x} = Ae^t$

- Blue line shows the exact solution
- Red dots show the euler approximation

1.2 Runge-Kutta second order method

The next method is called the Runge-Kutta second order method. This method takes a sort of weighted average of the slopes at the beginning and

the midpoint of the interval (Δt). Because of this, it will be more accurate than the Euler approximation, which only "refreshes" the slope every Δt . It works as follows.

- Choose a Δt
- Calculate $k_1 = \Delta t \cdot f(t_n, x_n)$ where $\dot{x} = f(t, x)$
- Calculate $k_2 = \Delta t \cdot f(t_n + \frac{\Delta t}{2}, x_n + \frac{k_1}{2})$
- Calculate $x_{n+1} = x_n + k_2$

This gives a better approximation of the integration of some inputted function, but it can still be improved. The result of this algorithm is shown below (for a Δt of 0.3) on the function $\dot{x} = x \cdot \sin t$.

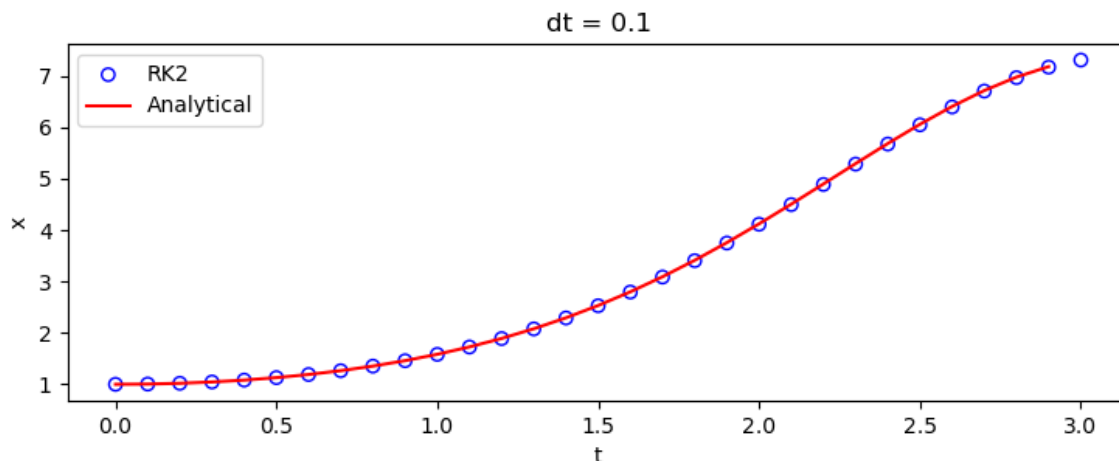


Figure 2: Runge-Kutta 2 estimate of $\dot{x} = x \cdot \sin t$

- Red dots show the exact solution
- Blue line shows the rk2 approximation

1.3 Runge-Kutta fourth order method

The final method is called the Runge-Kutta fourth order method. This method is similar to the Runge-Kutta second order method, but it takes the weighted average at four points instead of two. It works as follows.

- Choose a Δt
- Calculate $k_1 = f(t_n, x_n)$ where $\dot{x} = f(t, x)$
- Calculate $k_2 = f(t_n + \frac{\Delta t}{2}, x_n + \Delta t \cdot \frac{k_1}{2})$
- Calculate $k_3 = f(t_n + \frac{\Delta t}{2}, x_n + \Delta t \cdot \frac{k_2}{2})$
- Calculate $k_4 = f(t_n + \Delta t, x_n + \Delta t \cdot k_3)$
- Calculate $x_{n+1} = x_n + \frac{\Delta t}{6} \cdot (k_1 + 2k_2 + 2k_3 + k_4)$

Taking the weighted average of 4 points makes it very accurate, to the point where it is rarely necessary to go to a higher order Runge-Kutta. The result of the algorithm is shown below (for a Δt of 0.3) on the function $\dot{x} = x \cdot \sin t$.

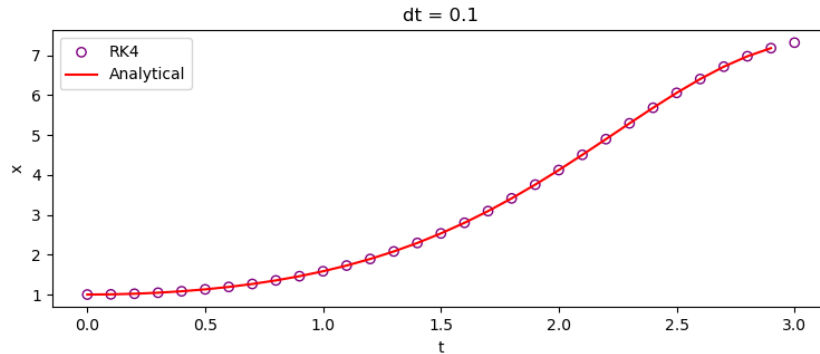


Figure 3: Runge-Kutta 4 estimate of $\dot{x} = x \cdot \sin t$

- Red dots show the exact solution
- Purple line shows the rk4 approximation

1.4 Comparing these methods

These three methods were introduced in order of increasing accuracy. The Euler method, with the slope calculated at one point, is the least accurate, and can be thought of as "Runge-Kutta first order". The Runge-Kutta second order method uses the slopes at two different points, and is more accurate. However, it still isn't good enough for more complicated functions. The Runge-Kutta fourth order method uses the slopes at four points, and is much more accurate than both of the other functions. To test this hypothesis, we test all of these methods on two different functions, and graph the error vs. the Δt . Here, error is calculated as the difference between the value of the actual function at $x = 3$ and the value estimated by each of the methods at $x = 3$.

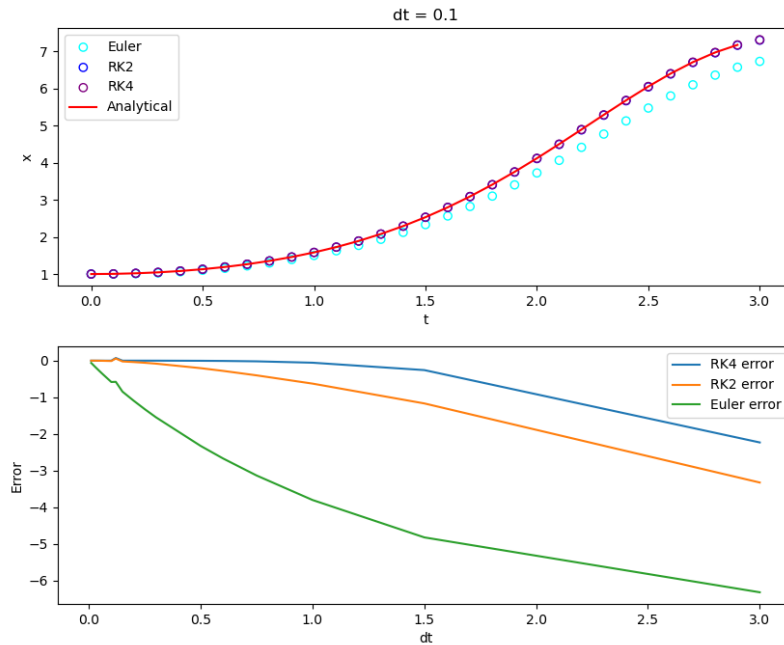


Figure 4: Various numerical integration methods on an integrable (separable) function, and error vs. Δt for those integration methods. Here, error is calculated as the difference between the value of the actual function at $x = 3$ and the value estimated by each of the methods at $x = 3$.

As shown in the graph, the error is 0 when Δt goes to 0. However, as Δt gets

larger, each approximation starts to deviate more from the actual solution. Additionally, the euler method deviates the most for higher values of Δt , and the RK4 method deviates the least. Now, we shall see what happens for an inseperable function. Here, we will use $\dot{x} = \sin(xt)$.

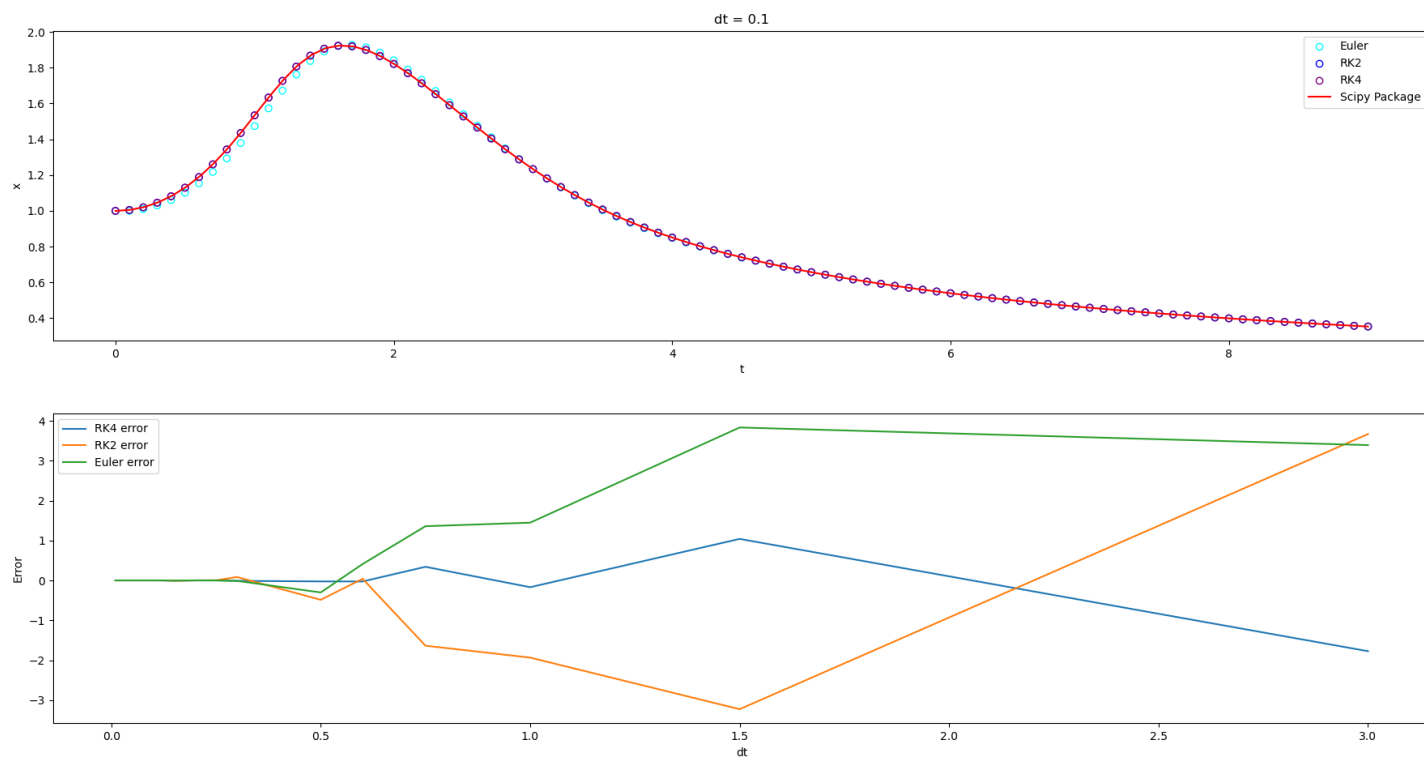


Figure 5: Various numerical integration methods on the function $\dot{x} = \sin(xt)$, and error vs. for those integration methods.

Since the integration cannot be calculated analytically, the "exact" solution was taken from the integration method of the python package SciPy. Again, we see that the RK4 method has the least error as Δt gets larger. This means that we can use the RK4 method for more complex equations.

1.5 Error of Euler method

In the Euler method, the approximation is given by:

$$\hat{x}(t + \Delta t) = x(t) + f(x, t)\Delta t \quad (1)$$

where $\dot{x} = f(x, t)$. The actual solution is given by:

$$x(t + \Delta t) = x(t) + \dot{x}(t)\Delta t + O(\Delta t^2) = x(t) + f(x, t)\Delta t + O(\Delta t^2) \quad (2)$$

The error is computed as:

$$x(t + \Delta t) - \hat{x}(t + \Delta t) = O(\Delta t^2) \quad (3)$$

1.6 Error of RK2 method

In the RK2 method, the approximation is given by:

$$\hat{x}(t + \Delta t) = x(t) + \frac{\Delta t}{2} (f(x, t) + f(x + \Delta t f(x, t), t + \Delta t)) \quad (4)$$

where $\dot{x} = f(x, t)$. We can now expand out the Taylor series for $f(x + \Delta t f(x, t), t + \Delta t)$:

$$f(x + \Delta t f(x, t), t + \Delta t) = f(x, t) + f_x(x, t)(\Delta t f(x, t)) + f_t(x, t)\Delta t + O(\Delta t^2) \quad (5)$$

With this, we can substitute it back into the RK2 definition:

$$\hat{x}(t + \Delta t) = x(t) + \frac{\Delta t}{2} (f(x, t) + f(x, t) + f_x(x, t)(\Delta t f(x, t)) + f_t(x, t)\Delta t + O(\Delta t^2)) \quad (6)$$

which becomes

$$\hat{x}(t + \Delta t) = x(t) + \Delta t f(x, t) + \frac{\Delta t^2}{2} f_x(x, t) f(x, t) + \frac{\Delta t^2}{2} f_t(x, t) + O(\Delta t^3) \quad (7)$$

The next step is to get the theoretical solution. We expand as:

$$x(t + \Delta t) = x(t) + \dot{x}(t)\Delta t + \frac{\ddot{x}(t)}{2}\Delta t^2 + O(\Delta t^3) \quad (8)$$

We have $\dot{x} = f(x, t)$, so we can compute:

$$\ddot{x} = \frac{\partial f(x, t)}{\partial t} + \frac{\partial f(x, t)}{\partial x} \cdot \frac{dx}{dt} = f_t(x, t) + f_x(x, t)f(x, t) \quad (9)$$

Substituting,

$$x(t + \Delta t) = x(t) + \Delta t f(x, t) + \frac{\Delta t^2}{2} f_x(x, t) f(x, t) + \frac{\Delta t^2}{2} f_t(x, t) + O(\Delta t^3) \quad (10)$$

Now, we can subtract to get the error.

$$x(t + \Delta t) - \hat{x}(t + \Delta t) = O(\Delta t^3) \quad (11)$$

Since the error of the Euler method is $O(\Delta t^2)$, and the error of the RK2 method is $O(\Delta t^3)$, the RK2 method will be much better than the Euler method for small values of Δt .

2 Motion of a spring

The motion of a spring can be written as a second order differential equation. In a spring, the force can be written as

$$F = -kx$$

where F is the force acting upon the spring, x is the displacement from the equilibrium position, and k is a constant (called the spring constant). Since $F = ma$ (where F is force, a is acceleration, and m is mass), and $a = \ddot{x}$, the equation of a spring can be written as:

$$\ddot{x} = -\omega^2 x, \text{ where } \omega = \sqrt{\frac{k}{m}}$$

This can be solved analytically. We start with

$$\frac{d^2 x}{dt^2} = -\omega^2 x$$

Then, we can substitute

$$\begin{aligned} \frac{dx}{dt} = v &\implies \frac{d^2 x}{dt^2} = \frac{dv}{dt} = \frac{dv}{dx} \cdot \frac{dx}{dt} = v \frac{dv}{dx} \\ v \frac{dv}{dx} &= -\omega^2 x \\ v dv &= -\omega^2 x dx \end{aligned}$$

We now have a first order differential equation.

$$\int v dv = \int -\omega^2 x dx$$

$$\frac{v^2}{2} = -\omega^2 \cdot \frac{x^2}{2} + c$$

By applying a boundary condition, we can get the value of c : $x = x_0 \implies v = 0$

$$0 = -\omega^2 \cdot \frac{x_0^2}{2} + c$$

$$c = \frac{\omega^2 x_0^2}{2}$$

Substituting it back in:

$$\frac{v^2}{2} = -\omega^2 \cdot \frac{x^2}{2} + \frac{\omega^2 x_0^2}{2}$$

$$v = \pm \omega \sqrt{x_0^2 - x^2}$$

We now write v in it's original form:

$$\frac{dx}{dt} = \pm \omega \sqrt{x_0^2 - x^2}$$

$$\pm \frac{dx}{\sqrt{x_0^2 - x^2}} = \omega dt$$

This differential equation could have two possible solutions:

$$\sin^{-1}\left(\frac{x}{x_0}\right) = \omega t \text{ or } \cos^{-1}\left(\frac{x}{x_0}\right) = \omega t$$

So we write the final answer as a linear combination of both:

$$x = Ax_0 \sin(\omega t) + Bx_0 \cos(\omega t)$$

Applying the first boundary condition, we can solve for B: At $t = 0$, $x = x_0$

$$B = 1$$

$$x = Ax_0 \sin(\omega t) + x_0 \cos(\omega t)$$

$$v = \frac{dx}{dt} = \omega Ax_0 \cos(\omega t) - \omega x_0 \sin(\omega t)$$

Applying the second boundary condition, we can solve for A: At $t = 0$, $v = 0$

$$0 = \omega A \implies A = 0$$

Finally, we get:

$$x = x_0 \cos(\omega t)$$

For a numerical solution, it is best to write the equation of motion in a vector form. Let

$$\frac{d\vec{y}}{dt} = \vec{f}(\vec{y}) = \begin{pmatrix} -\omega^2 y_2 \\ y_1 \end{pmatrix} \quad (12)$$

Now, we can use this modified version of the RK4 method:

$$\begin{aligned} \vec{k}_1 &= \vec{f}(\vec{y}_n) \\ \vec{k}_2 &= \vec{f}(\vec{y}_n + \frac{h}{2} \cdot \vec{k}_1) \\ \vec{k}_3 &= \vec{f}(\vec{y}_n + \frac{h}{2} \cdot \vec{k}_2) \\ \vec{k}_4 &= \vec{f}(\vec{y}_n + h \cdot \vec{k}_3) \\ \vec{y}_{n+1} &= \vec{y}_n + \frac{h}{6} (\vec{k}_1 + 2\vec{k}_2 + 2\vec{k}_3 + \vec{k}_4) \end{aligned}$$

Here, $\vec{y}_n^{(i)}$ represents the i th component of $\vec{y}$ at the timestep n . The results of using this on the equation of a spring are shown below:

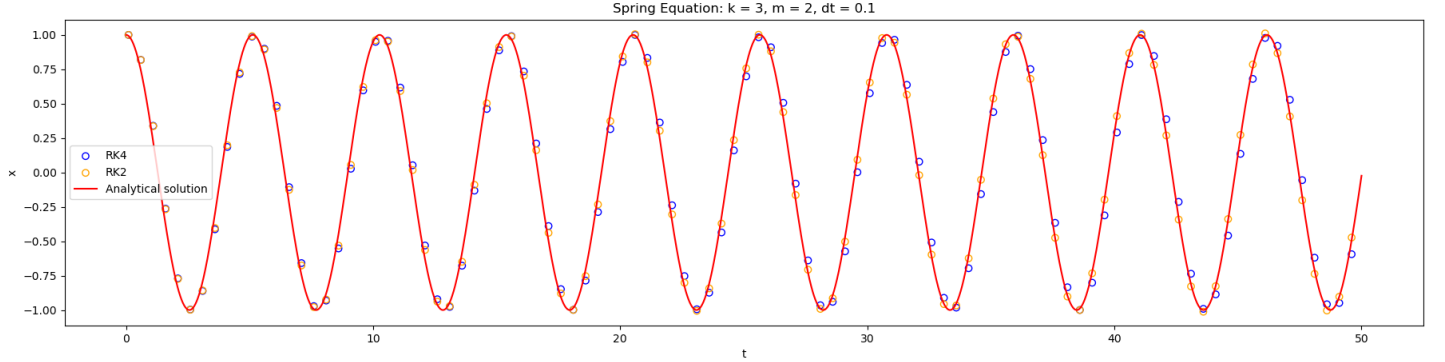


Figure 6: Runge-Kutta 4, 2 estimate of the motion of a spring
- Red dots show the exact (analytical) solution, Blue line shows the RK4 approximation, Orange line shows the RK2 approximation. Here, $h = 3$ and $m = 2$, which are the spring constant and the mass respectively.

To once again show the difference between the RK4 and RK2 approximations,

we create a graph showing the error (at the final point) for both methods vs. Δt .

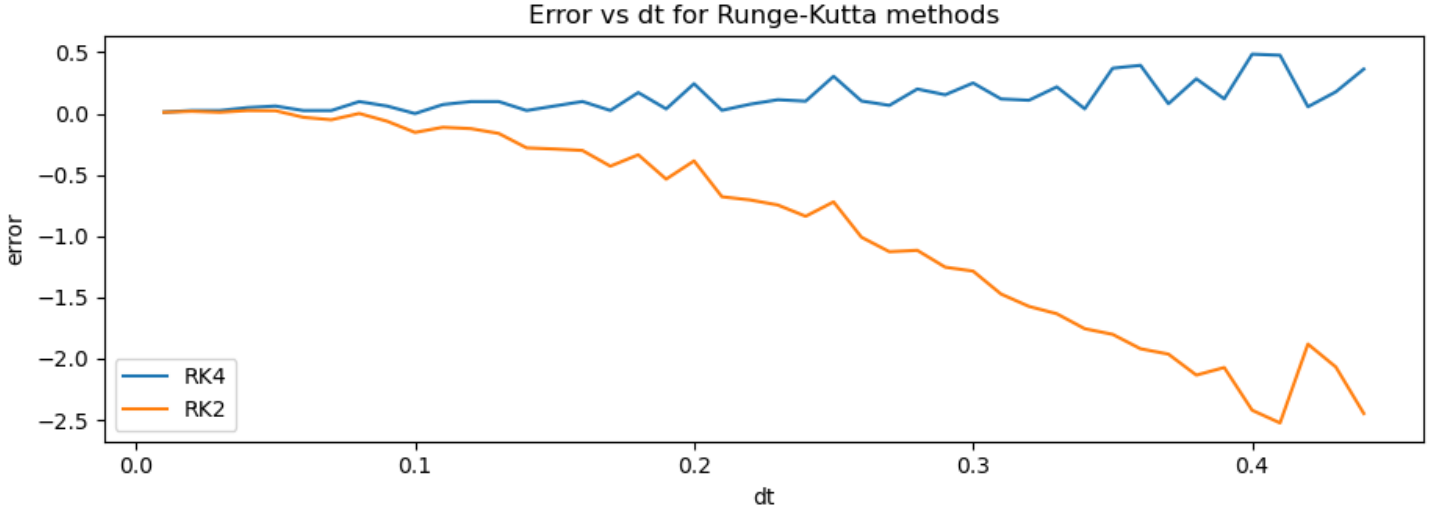


Figure 7: Error of Runge-Kutta 4, 2 for the motion of a spring
- Blue line shows the error of the RK4 approximation for various dts at $t = 50$, Orange line shows the error of the RK2 approximation for various dts at $t = 50$.

We can also apply the same strategy on the equation of a damped spring.

$$\vec{y} = \begin{pmatrix} v \\ x \end{pmatrix} \quad (13)$$

Numerical Spring Equation:

$$\frac{d\vec{y}}{dt} = \vec{f}(\vec{y}) = \begin{pmatrix} -\omega^2 y_1 - \alpha y_0 \\ y_0 \end{pmatrix} \quad (14)$$

Analytical Spring Equation:

$$\begin{aligned} \frac{d^2 x}{dt^2} + \alpha \frac{dx}{dt} + \omega^2 x &= 0 \\ (D^2 + \alpha D + \omega^2)x &= 0 \end{aligned}$$

Roots of the Characteristic equation:

$$r_1, r_2 = \frac{-\alpha \pm \sqrt{\alpha^2 - 4\omega^2}}{2}$$

Which implies:

$$x(t) = C_1 \exp(r_1 t) + C_2 \exp(r_2 t) \quad (15)$$

Boundary conditions:

$$\begin{aligned} x_0 &= C_1 + C_2 \\ 0 &= C_1 r_1 + C_2 r_2 \end{aligned}$$

Implies:

$$\boxed{x(t) = \frac{r_2 x_0}{r_2 - r_1} \exp(r_1 t) + \frac{r_1 x_0}{r_1 - r_2} \exp(r_2 t)} \quad (16)$$

Curiously, the sum of two exponentials created a damped sine graph. However, this can be explained by using Euler's formula for complex numbers. When $\alpha < 4\omega^2$, r_1 and r_2 become complex numbers. Let

$$c = \frac{\sqrt{4\omega^2 - \alpha^2}}{2},$$

so we can write

$$r_1, r_2 = \frac{-\alpha}{2} \pm ci.$$

Now, $x(t)$ can be written as:

$$x(t) = \frac{r_2 x_0}{r_2 - r_1} \cdot e^{-\frac{\alpha t}{2}} \cdot e^{cti} + \frac{r_1 x_0}{r_1 - r_2} \cdot e^{-\frac{\alpha t}{2}} \cdot e^{-cti}$$

Expanding out e^{cti} , we can see where the sine shape comes from (which is also scaled by an exponential function, causing the damping)

$$x(t) = \Re(e^{-\frac{\alpha t}{2}} \cdot (\frac{r_2 x_0}{r_2 - r_1} \cdot (\cos(ct) + i \sin(ct)) + \frac{r_1 x_0}{r_1 - r_2} \cdot (\cos(ct) - i \sin(ct))))$$

However, if $\alpha > 4\omega^2$, r_1 and r_2 are real. In this case, the system is damped enough such that there is no oscillation at all. An example is shown below:

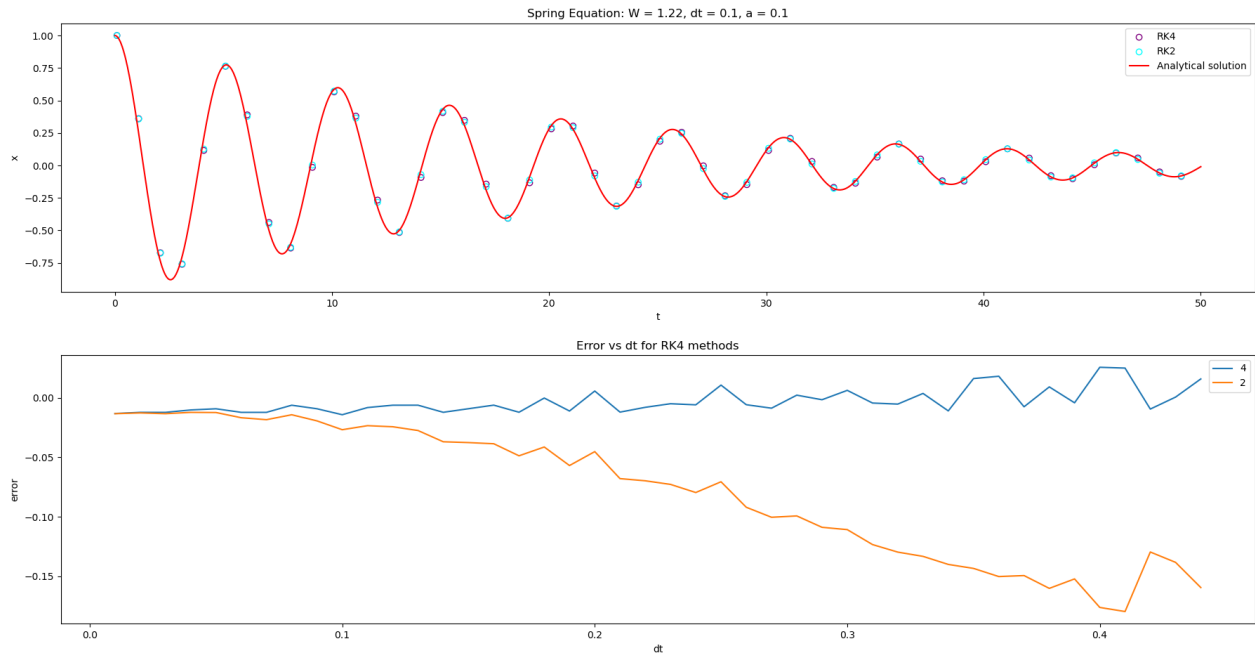


Figure 8: The first picture shows the analytical solution vs. the rk2 and rk4 approximations. The second picture shows the error of the rk2 and rk4 approximations vs the analytical solution.

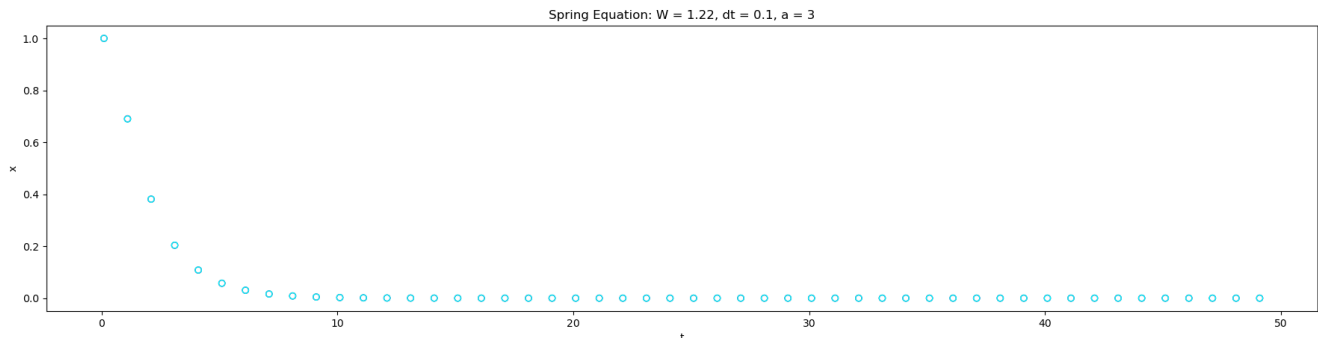


Figure 9: A spring system with a large α value: This is called an over-damped/critically damped spring.

3 Double Pendulum

The double pendulum is one of the simplest systems that exhibit chaotic behavior. This means that slight changes in the initial position will cause

massive differences over time. A double pendulum consists of two rods with equal lengths and masses. The rods have their mass distributed equally across it. There are four components involved in the motion of the double pendulum. θ_1, θ_2 refer to its position (angle between the rod and the vertical), and $\dot{p}_{\theta_1}, \dot{p}_{\theta_2}$ refer to the momentum of each rod.

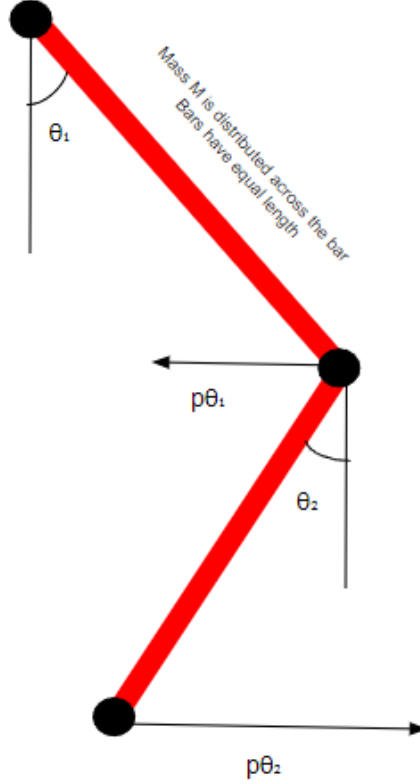


Figure 10: Double Pendulum Sketch: m is the mass of each bar, and l is the length of each bar.

The equation of the double pendulum is as follows:

$$\frac{d\vec{y}}{dt} = \begin{pmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{p}_{\theta_1} \\ \dot{p}_{\theta_2} \end{pmatrix} = \vec{f}(\vec{y}) = \begin{pmatrix} \frac{6}{ml^2} \frac{2p_{\theta_1} - 3\cos(\theta_1 - \theta_2)p_{\theta_2}}{16 - 9\cos^2(\theta_1 - \theta_2)} \\ \frac{6}{ml^2} \frac{8p_{\theta_2} - 3\cos(\theta_1 - \theta_2)p_{\theta_1}}{16 - 9\cos^2(\theta_1 - \theta_2)} \\ -\frac{1}{2}ml^2 \left(\dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + 3\frac{g}{l} \sin \theta_1 \right) \\ -\frac{1}{2}ml^2 \left(-\dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) + \frac{g}{l} \sin \theta_2 \right) \end{pmatrix} \quad (17)$$

In this case, the differential equations cannot be solved analytically, so it is necessary to solve it numerically.

We can use the same method of vectorizing the equations, and using the Runge-Kutta 4 method.

To see the results, we can graph θ_1 vs $\dot{p}_{\theta_1}$ and θ_1 vs $\dot{p}_{\theta_1}$ to see a 2D slice of the 4D space created by the double pendulum. We can also graph θ_1 vs. θ_2 to get further insight about the system. Both pictures are shown below:

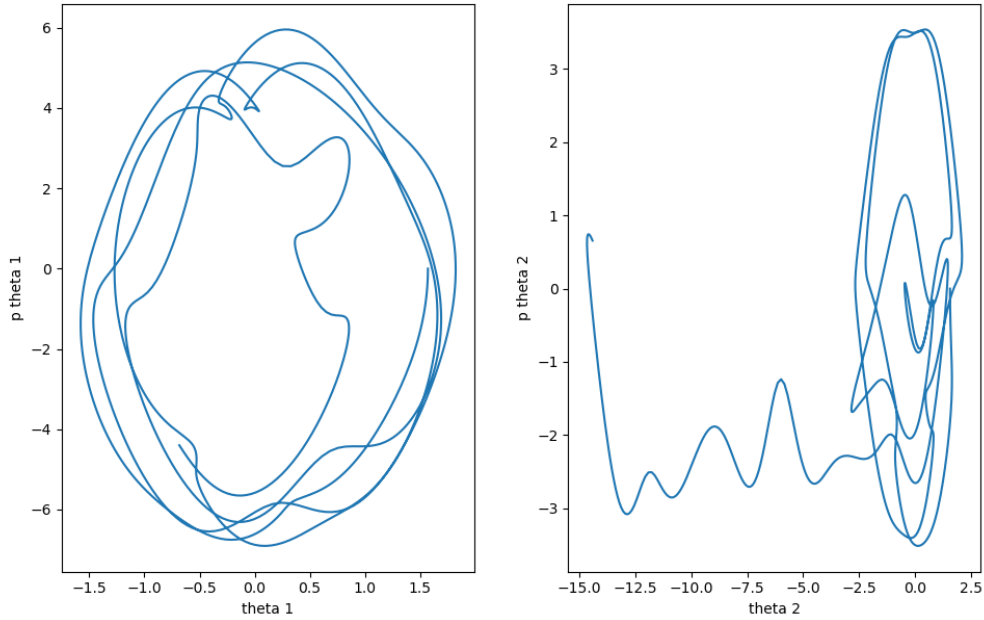


Figure 11: θ_1 vs $\dot{p}_{\theta_1}$ and θ_1 vs $\dot{p}_{\theta_1}$ for the starting values $\theta_1 = \theta_2 = \frac{\pi}{2}$, $\dot{p}_{\theta_1} = \dot{p}_{\theta_2} = 0$

One interesting thing to observe in a double pendulum is whether it "flips" or not. A flip refers to whether or not θ_2 every becomes greater than π or less than $-\pi$. In this paper, a 600x600 grid of starting values θ_1, θ_2 from $-\pi, \pi$ will be analyzed to check for flips. We will check for flips within 50 seconds. However, in order to get an accurate picture, we first need to make sure that the data we collect is close enough to the actual result. We can do this by creating a series of Δt , where each Δt is half of the previous. Then, we can

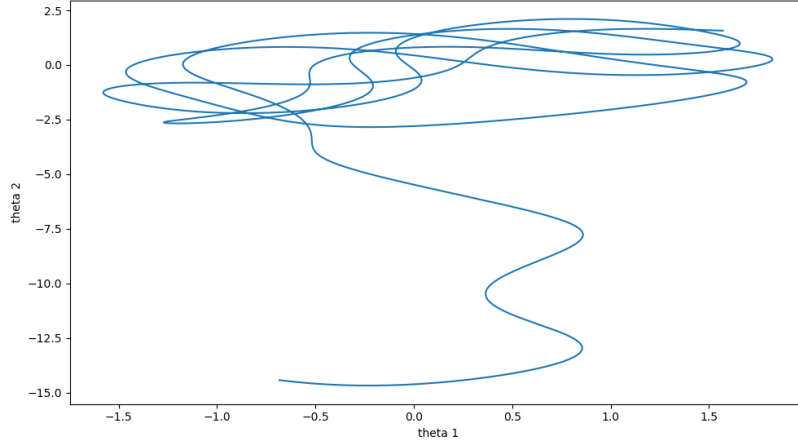


Figure 12: θ_1 vs θ_2 for the starting values $\theta_1 = \theta_2 = \frac{\pi}{2}, \dot{\theta}_1 = \dot{\theta}_2 = 0$

compare the end position of the smaller Δt with the bigger one, and see how much the difference is (at time = 50). When the distance goes below a certain value (0.05 in this case), we can say that anything below that Δt will result in a difference less than 0.05, meaning that the end result is close enough to a theoretical simulation in which $\Delta t = 0$.

Different values of θ_1, θ_2 may require different values of Δt , so the appropriate Δt for every one of the points in the 600x600 grid is found. The result of this is shown below:

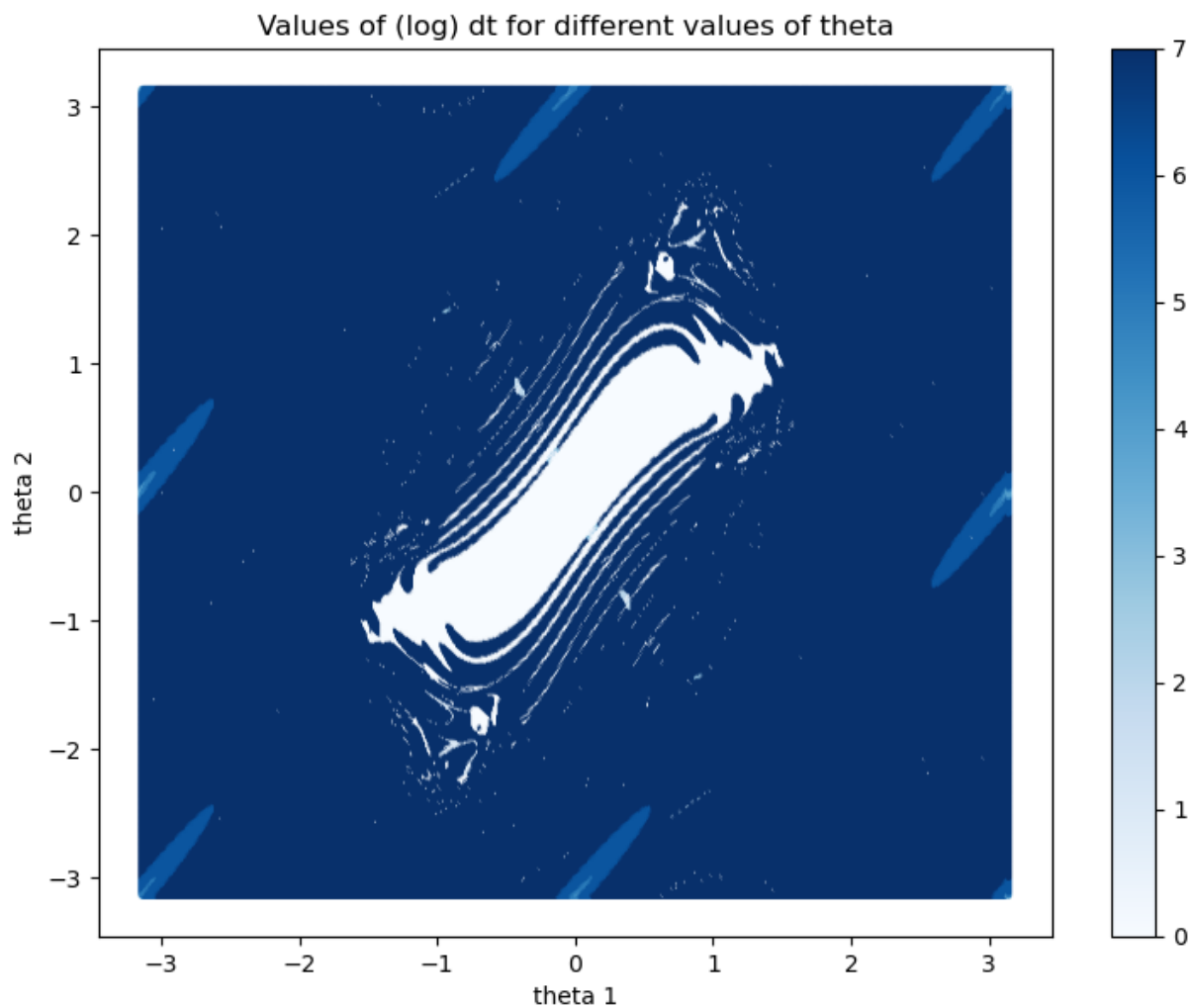


Figure 13: Δt required for each point: Each point is assigned a number i , where the Δt for that point can be evaluated as $0.1 \cdot 0.5^i$. The minimum Δt required was $0.1 \cdot 0.5^7$, or 0.00078125.

Now, we can find the results of which starting values of θ_1, θ_2 cause the double pendulum to flip. The graph is shown below.

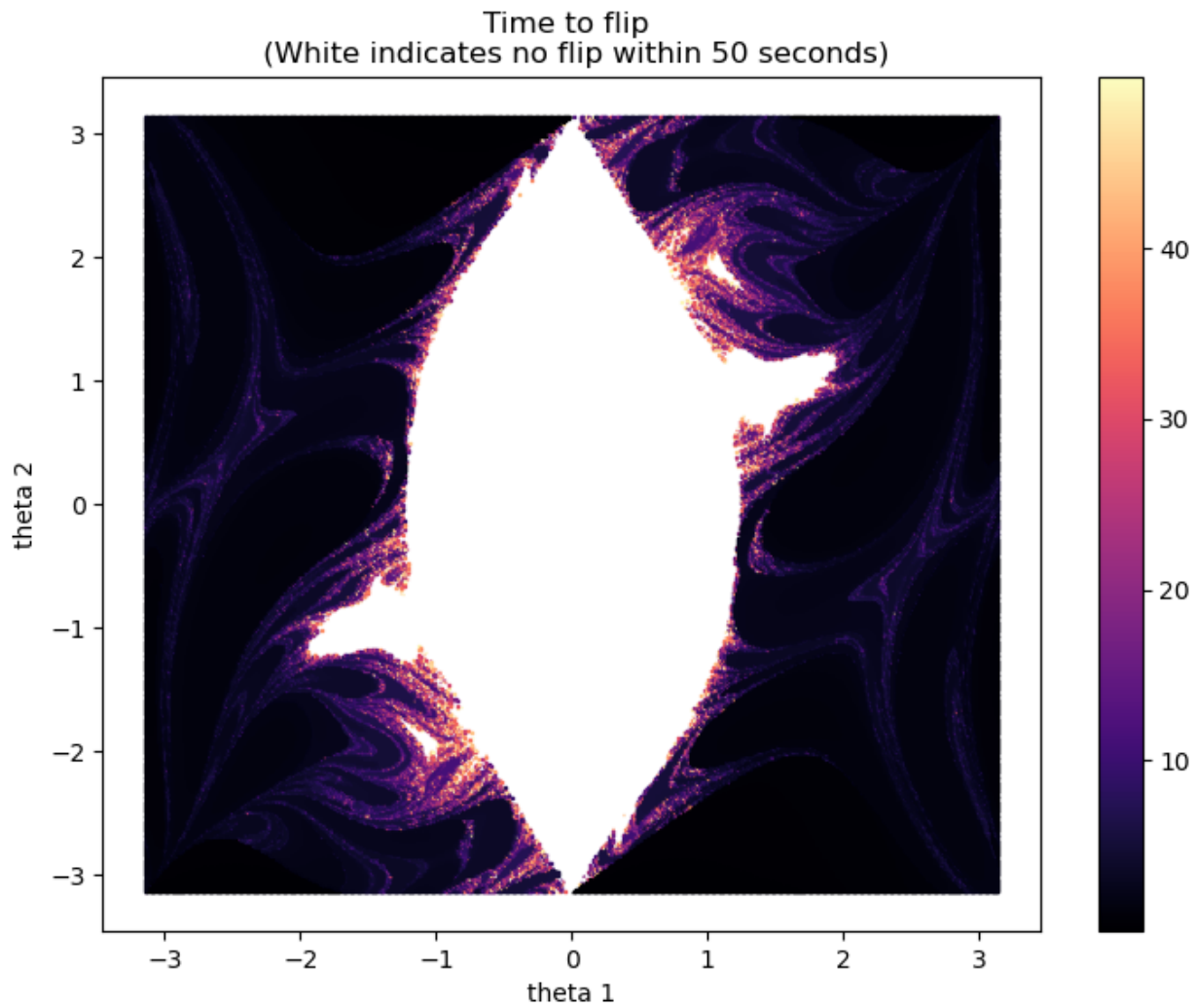


Figure 14: Results for flips in the double pendulum for different starting values of θ_1, θ_2 . The figure is color coded by time taken to flip, where white represents no flip within 50 seconds.

4 Analysis

Figure 11 shows the Δt required for different values of θ_1, θ_2 . As expected, values of θ_1, θ_2 closer to 0 don't need very small values of Δt , because there is not a lot of movement in the first place. Interestingly, there are regions on the edges that don't need the smallest value of Δt . However, most of the graph does need to go to $i = 7$ in order to be accurate enough for the analysis.

In figure 12, there is a large white eye-shaped space in the middle, which represents the starting values of θ_1, θ_2 that didn't cause the double pendulum to flip. This is most likely because for those values closer to 0, there was not enough energy in the system for a flip to occur. On the borders of that shape, higher colors can be seen, because they took a long time to flip. However as more energy gets introduced into the system (through greater values of θ_1, θ_2), the flip happens faster and faster - shown by the black close to the borders of the figure. Purple swirls can also be seen in that area, representing values of θ_1, θ_2 which took slightly longer to flip.

Another interesting part of the Figure 12 are the strange white appendages sticking out of the main eye shape on the bottom right and top left. In these regions, the double pendulum doesn't flip, even though there should be enough energy to make it happen.